MIL-T-60699(MU)
22 January 1969

SUPERSEDING
FA-PD-MI-2691
15 September 1965

MILITARY SPECIFICATION
THRUSTER, CARTRIDGE ACTUATED: M7
REMANUFACTURED METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Thruster, Cartridge Actuated: M7 Remanufactured Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-171	- Finishing of Metal and Wood
MIL-STD-1167	- Ammunition Data Card

DRAWINGS

Munitions Command

F8593278	- Thruster, Cartridge Actuated, M7, Metal Parts Assembly
C8796480	- Ammunition Container, Fiber, M378 for Thruster, Cartridge Actuated, M7

FSC 1377

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DRAWINGS (Cont'd)

C8796479 - Box, Packing, Ammunition for Thruster, Cartridge Actuated, M7, in Fiber Container M378
IEL 8593278 - Inspection Equipment List for Thruster, Cartridge Actuated, M7, Metal Parts Assembly

(Copies of specifications, standards and drawings required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Material.- Components for remanufacture shall be obtained from the M7 Thruster model. All components from earlier model types shall be identified and segregated for required modification prior to assembly (see 6.4.2).

3.3 Hydrostatic pressure.- The body and head shall withstand a constant internal pressure of 5000 pounds per square inch (psi) minimum, for 15 seconds, minimum, without application of any external load and shall show no evidence of leakage or mechanical failure (see 6.2).

3.4 Structural strength.- The assembled thruster shall withstand a static tensile force of 500 pounds, minimum for 15 seconds, minimum, between mounting points without unlocking and without mechanical failure. (See 6.2).

3.5 Metal parts assembly.- Parts shall be combined to produce an assembly which conforms to the requirements of the metal part assembly drawing, F8593278. All threaded joints shall be assembled with sufficient torque to tightly bottom components.

3.5.1 The metal parts assembly shall be assembled with the following new metal part components: Shear Pin 8593284; Piston Locking Spring, 8593287; O-Rings; and Lubricant.

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3.5.2 The metal parts assembly shall include component parts that comply with the following drawing requirements: Head, F8593279, with setscrew threads #10 (.190)-32NF-3B; Body, D8593280, with thread lead - 5/16 long; Guide, C8593282, with location of shear pin hole - .193 ± .005; Firing Pin, D8593864, with striker length -.229 + .003. Any components that do not comply with above drawing requirements shall be either modified or discarded. Set screws or any thread component which shows evidence of cross threading, stripping or excessive wear shall be replaced (see 6.4.2).

3.5.3 End sleeve rotation.- Upon completion of assembly and after structural strength test, the end sleeve shall be free to rotate 360 degrees in a clockwise direction.

3.6 Protective finish.- Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour configuration. Remanufactured exterior components shall have the existing protective finish stripped and the specified protective finish reapplied (see 6.4.1)

3.7 Residual magnetism.- The metal parts assembly shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches (see 6.3).

3.8 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.8.1 Metal defects.- All components shall be free from cracks, splits, or other metal defects.

3.8.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define Quality Assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550 and as specified herein.

4.2 Inspection provisions.-

4.2.1 Lot.

4.2.1.1 Submission of product.- Unless otherwise specified by the contracting officer, formation of lots and presentation of lots shall be in accordance with the "Submission of Product" provisions of MIL-STD-105.

4.2.1.2 Size of lot.- The shipping lots of metal parts assemblies shall consist of 1200 thrusters, maximum, provided the homogeneity of remanufacturing process has been retained throughout the inspection lot. A smaller lot may be shipped upon approval of the contracting officer.

4.2.2 Examination.- The roughness comparison specimens prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

4.2.2.1 Reclaimed parts.- Examination of reuseable parts, which require no reprocessing, shall be limited to inspection of defects that require visual type inspection. One hundred per-cent examination shall be performed for all such defects. All nonconforming parts shall be either reprocessed or rejected.

4.2.2.2 Reprocessed parts.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table I and Table II. For an individual defect (single characteristic) the criteria in Table I shall be used. For total defects within a major or minor group, the criteria in Table II shall be used. All nonconforming material shall be rejected.

Table I

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

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Table II

Total Allowable Defects per Sample Size

Sample Sizes for Major Defects Number of Defects in Class	110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14

Sample Sizes for Minor Defects Number of Defects in Class	110	150	225	300
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

4.2.2.3 Head, Drawing F8593279 covering a detail of Drawing F8593278

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Set screw thread	Gage
102.	Thread damaged or not full	Visual
103.	Metal defective	Visual
104.	Protective finish inadequate or defective	Visual
105.	Burr	Visual
106.	Foreign matter	Visual
MINOR		
	None defined	

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4.2.2.4 Body, Drawing D8593280 covering a detail of Drawing F8593278

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR		
101.	Length of thread lead	Gage
102.	Thread damaged or not full	Visual
103.	Metal defective	Visual
104.	Protective finish inadequate or defective	Visual
105.	Burr or sharp corner	Visual
106.	Foreign matter	Visual
MINOR		
None defined		

4.2.2.5 Sleeve, End, Drawing D8593281 covering a detail of Drawing F8593278

CRITICAL		
None defined		
MAJOR		
101.	Thread damaged or not full	Visual
102.	Metal defective	Visual
103.	Protective finish inadequate or defective	Visual
104.	Burr or sharp corner	Visual
105.	Foreign matter	Visual
MINOR		
None defined		

4.2.2.6 Guide, Drawing 8593282 covering a detail of Drawing F8593278

CRITICAL		
None defined		

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4.2.2.6 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Location of shear pin hole	Gage
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr or sharp corner	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.7 Pin, Firing, Drawing D8593864 covering a detail of Drawing F8593278

CRITICAL	
None defined	
MAJOR	
101. Length of striker	Gage
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr or sharp corner	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.8 Piston, Drawing D8593285 covering a detail of Drawing F8593278.

CRITICAL	
None defined	
MAJOR	
101. Thread damaged or not full	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr or sharp corner	Visual
105. Foreign matter	Visual

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4.2.2.8 (Cont'd)

Defect	Method of Inspection

MINOR

None defined

4.2.2.9 Collar, Drawing C8593286, covering a detail of Drawing F8593278.

CRITICAL

None defined

MAJOR

101.	Metal defective	Visual
102.	Protective finish inadequate or defective	Visual
103.	Burr or sharp corner	Visual
104.	Foreign matter	Visual

MINOR

None defined

4.2.2.10 Key, Drawing 8593288 covering a detail of Drawing F8593278

CRITICAL

None defined

MAJOR

101.	Metal defective	Visual
102.	Protective finish inadequate or defective	Visual
103.	Burr or sharp corner	Visual
104.	Foreign matter	Visual

MINOR

None defined.

4.2.2.11 Metal Parts Assembly, Drawing F8593278 less all components except Firing Pin 8593864, Guide 8593282, and O-ring MS28775-012.

CRITICAL

1.	Protrusion of firing pin	Gage

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4.2.2.11 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Any part missing	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.12 Metal Parts Assembly, Drawing F8593278.

CRITICAL	
None defined	
MAJOR	
101. End sleeve not free to rotate 360 degrees	Manual
102. Any part missing, except as otherwise classified	Visual
103. Improper assembly	Visual
104. Thread damaged or not full	Visual
105. Marking missing, incorrect or illegible	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR	
201. Shipping cap or plug missing	Visual
202. Protective finish inadequate or defective	Visual

4.2.2.13 Container, Fiber, Drawing C8796480.

CRITICAL	
None defined	
MAJOR	
None defined	

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4.2.2.13 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual

4.2.2.14 Box, Packing, Drawing C8796479.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual & Visual
203. Strapping missing, loose, broken or improperly assembled	Manual & Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking misleading or unidentifiable	Visual
207. Number of units incorrect	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with provisions of 4.3.

4.2.3.1 Hydrostatic pressure.- Each body and each head shall be subjected to an internal hydrostatic pressure. Failure of any unit to comply with the requirements of 3.3 shall be cause for rejection of the unit.

4.2.3.2 Structural strength.- Each thruster shall be subjected to the structural strength test. Failure of any thruster to comply with the requirements of 3.4 shall be cause for rejection of the thruster.

4.2.3.3 Protective finish.- The number of units for protective finish tests shall be in accordance with the applicable MIL-Standard. Failure of any sample to comply with the requirements shall be cause for rejection of the lot.

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4.2.3.4 Residual magnetism.- Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any unit to comply with the requirements of 3.7 shall be cause for rejection of the assembly.

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-8593278.

4.3 Test methods and procedures.

4.3.1 Hydrostatic pressure.- The body and head, assembled or individually shall be mounted in an approved device and subjected to an internal hydrostatic pressure in accordance with the requirements of 3.3. The testing device used shall internally seal the body and end cap, as applicable, at each end without applying any axial load other than that applied by liquid pressure.

4.3.2 Structural strength.- The thruster shall be subjected to the structural strength test in accordance with the requirements of 3.4. The test shall be performed using a fixture in conjunction with a tensile testing machine or other device.

4.3.3 Protective finish.- Test procedures for protective finish shall be in accordance with the specification referenced on the component drawing.

4.3.4 Residual magnetism.- The thruster shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the thruster 6 inches from and at the same height as an approved compass, in a north-south horizontal position (head end at south) with the compass set in an east-west heading. The thruster shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the thruster.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings C8796480 and C8796479.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

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6.2 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 When remanufacturing and loading of assembly is performed by the same facility, the requirement of 3.7 is waived and the residual magnetism performed on the loaded assembly only.

6.4 Special instructions covering the modification, repair and assembly of remanufactured components shall be as follows:

6.4.1 Protective finish.- Remanufactured components which include Head, Body, and End Sleeve shall be stripped of protective finish and replated in accordance with the applicable drawings and specifications.

6.4.2 Modification.

6.4.2.1 Head, 8593279, modify all setscrew threads to #10(.190)-32NF-3B, Metal stamp or engrave an arrow as specified in Detail D of drawing F8593279.

6.4.2.2 Body, 8593280 modify thread lead to be 5/16 long.

6.4.2.3 Guide, 8593282, locate shear pin hole .193 ± .005 from end. All holes at different location shall be plugged.

6.4.2.4 Firing Pin, 8593864, modify striker length to .229 + .003.

6.5 Supersession data.- This specification includes the requirements of Purchase Description FA-PD-MI-2691 dated 15 September 1965.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A197

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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